



Birla Institute of Technology & Science, Pilani
Pilani | Dubai | Goa | Hyderabad
ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27
Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : BITS F343
Course Title : FUZZY LOGIC & APPLICATIONS
Instructor-in-charge : CHANDRA SHEKHAR
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Chamber No.: 6067-E
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1. Course description:

Fuzzy sets, fuzzy binary relations, fuzzy logic, fuzzy reasoning, applications in decision making, control theory, expert systems, artificial intelligence etc.

2. Scope and Objective:

The course aims to provide a comprehensive understanding of the fundamental concepts of fuzzy sets, fuzzy relations, fuzzy logic, and fuzzy reasoning, along with their applications in decision making, control theory, expert systems, artificial intelligence, and other interdisciplinary areas. Emphasis is placed on developing analytical, computational, and problem-solving skills through real-world applications. Each student is required to undertake a project involving the application of fuzzy logic, which may be study-oriented, computer-oriented, simulation-based, or model-based, and provides an opportunity to explore research articles and recent developments in fuzzy logic and its applications.

3. Course learning outcomes:

CLO1. Explain the fundamental concepts and mathematical foundations of fuzzy sets, fuzzy numbers, fuzzy relations, and fuzzy logic.

CLO2. Formulate and analyze problems involving fuzzy relations, fuzzy reasoning, fuzzy inference, and related logical structures.

CLO3. Apply fuzzy logic techniques to real-world problems in decision making, control systems, expert systems, artificial intelligence, and other interdisciplinary areas.

CLO4. Develop and evaluate a fuzzy logic-based project using appropriate computational tools, with reference to relevant research literature and contemporary applications.

4. Text books and references:

TB1. M. Ganesh: Introduction to Fuzzy Sets and Fuzzy Logic, PHI, (2006)

RB1. George. J. Klir & Bo Yuan: Fuzzy Sets and Fuzzy Logic, PHI, 1997.

RB2. Timothy J. Ross: Fuzzy Logic with Engineering Applications, John Wiley & Sons, 2004.

RB3. Kwang H. Lee: First Course on Fuzzy Theory and Applications: Springer International Edition (2005).

5. Course Format & Weekly Workflow

weekly schedule for 14 weeks with [3] lectures per week

Wk no.	Module no.	Description	Ref.	Learning Outcome
1	Module 1: Fuzzy Sets and Applications	Basic concepts, set operations, properties, and representation of crisp sets.	Ch.1	Understand and apply fundamental concepts and operations of crisp sets.
2		Introduction to fuzzy sets, membership functions, representations, operations, and basic properties.	Ch.6	Explain and analyze fuzzy sets using membership functions and set operations.
3		Advanced concepts of fuzzy sets, α -cuts, support, core, height, normality, convexity, and related properties.	Ch. 6	Analyze structural properties of fuzzy sets and interpret their practical significance.
4	Module 2: Fuzzy Numbers	Definition, types, representation, arithmetic operations, and applications of fuzzy numbers.	RB1	Represent and perform basic operations on fuzzy numbers.
5	Module 3: Fuzzy Relations	Fuzzy binary relations, representation, properties, composition, and related operations.	Ch. 7	Formulate, represent, and analyze fuzzy binary relations.
6	Module 4: Fuzzy Logic	Propositions, logical connectives, truth tables, predicates, quantifiers, and logical structures.	Ch. 2 & 3	Apply principles of propositional and predicate logic to logical problems.
7		Fuzzy propositions, linguistic variables, fuzzy connectives, implications, and basic fuzzy logic concepts.	Ch. 8	Explain and formulate logical relationships in a fuzzy environment.
8		Fuzzy rules, inference mechanisms, fuzzy reasoning, and rule-based systems.	Ch. 8	Apply fuzzy reasoning and inference techniques to decision problems.
9	Module 5: Switching Functions and Boolean Algebra	Boolean variables, Boolean functions, switching functions, Boolean operations, and fundamental laws.	Ch. 4 & 5	Apply Boolean algebra and switching functions to logical modeling.
10	Module 6: Applications of Fuzzy Logic	Fuzzy decision making, control systems, expert systems, and artificial intelligence.	Ch. 9 & 10	Apply fuzzy logic concepts to interdisciplinary real-world applications.
11		Application-oriented modeling and analysis of fuzzy systems.	Ch. 9 & 10	Formulate and analyze application-oriented fuzzy models.
12	Module 7: Fuzzy Probability	: Basic concepts, formulation, interpretation, and applications of fuzzy probability.	RB3	Explain and apply fundamental concepts of fuzzy probability.
13	Module 8: Fundamentals of Neural Networks	Basic concepts, artificial neurons, network structures, and learning principles.	RB4	Explain the fundamental structure and working principles of neural networks.
14	Module 9: Integrated Applications and Project Work	Integration of fuzzy logic concepts through applications, computational exercises, and project-oriented work.	RB1- RB4	Integrate fuzzy concepts to formulate, implement, and analyze an application-oriented problem.

6. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Assignment/Presentation	10 %		TBA	
Announced Quizzes	20 % (Max. Marks 20)	20 min, Marks 10 for each Best 2 out of 3	TBA	Closed book
Mid Semester	30 % (Max. Marks 30)	90 min	5/10/26 FN1	Closed book
Comprehensive	40 % (Max Marks: 40)	180 min	11/12/26 FN	Open book/

Criterion for NC: Less than 10% of maximum marks.

Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

7. **Chamber consultation hours:** Th-7 (Chamber No.: 6067-E)

8. **Makeup Policy:** [Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.]



Instructor-in-charge

[BITS F343]